



Mirjabbar Badalov (ID: 285269)

Substations and Lines (MPA-ESV 25/26L)

Laboratory Report A (17.02.2026)

Parallel operation of two transformers

1. Introduction and Objective

The objective of this laboratory exercise was to verify the feasibility and conditions for the parallel operation of two transformers, and to determine the maximum power they can transfer without overloading either unit. Additionally, the experiment aimed to analyse the load distribution between the individual transformers when supplying a common load.

During the lab, a single-phase connection was established. The setup included two parallel-connected transformers supplying a resistive load, with ammeters placed to measure the primary current of each transformer as well as the total current. A voltmeter was used to measure the voltage across the common load. By adjusting the load resistance, measurements were taken at three different operating points.

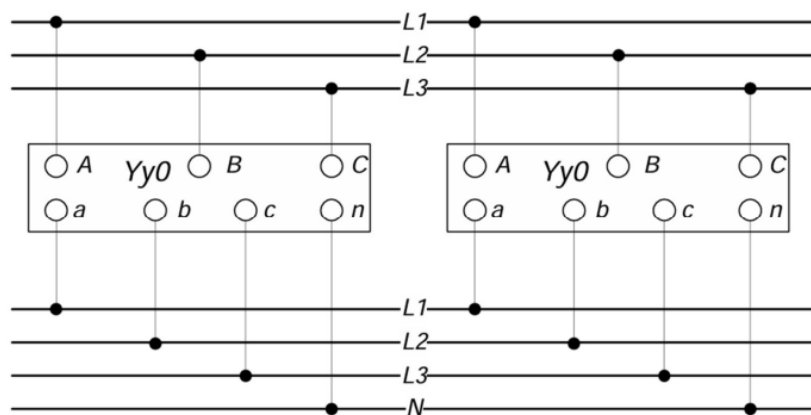


Figure 1 Parallel operation of two 2300 VA Yy0 transformers

2. Evaluation of Conditions for Parallel Operation

Before placing the transformers in parallel, we evaluated their nameplate parameters against the theoretical conditions required for successful parallel operation (as outlined in section 1.2.4 of the lab instructions).

Transformer 1 (TR1) Parameters:

- Rated Apparent Power (S_{nl}): 2300 VA
- Primary/Secondary Voltage: 3 * 400 V / 3 * 230 V
- Short-Circuit Voltage (u_{kl}): 3.9%
- Vector Group: Yy0

Transformer 2 (TR2) Parameters:

- Rated Apparent Power (S_{nl}): 1100 VA
- Primary/Secondary Voltage: 3 * 400 V / 3 * 230 V
- Short-Circuit Voltage (u_{kl}): 4.23%
- Vector Group: Yy0

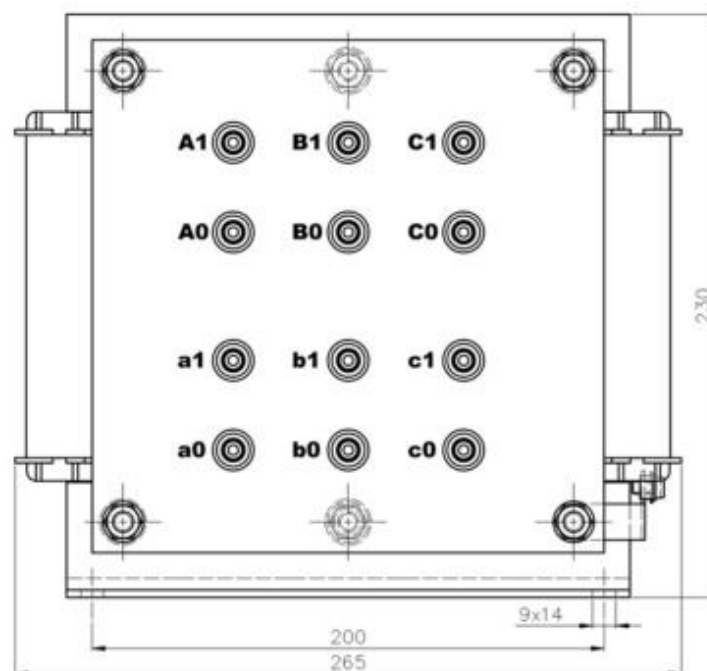


Figure 2 2300 VA transformer used in laboratory

Analysis of the Conditions: Even though the transformers are 3-phase devices by design, we connected them as a 1-phase system. The evaluation of the theoretical conditions is as follows:

1. **Identical phase sequence:** In our 1-phase connection, this translates to ensuring correct polarity, which was observed to prevent a direct short circuit.
2. **Identical phase shift (same vector group):** Both transformers have a Yy0 vector group, satisfying this condition.
3. **Identical transformation ratio:** Both transformers have a ratio of 400 V / 200 V ($p \approx 1.739$). This condition is perfectly met, preventing significant circulating currents.
4. **Power ratio limit (no more than 2:1):** The ratio of the rated powers is $S_{n1} / S_{n2} = 2300 / 1100 \approx 2.09$. This slightly exceeds the recommended 2:1 ratio limit. While not strictly ideal, it is close enough to function safely in a controlled laboratory environment.
5. **Short-circuit voltage relationship:** The transformer with the lower power rating should have a slightly higher short-circuit voltage. This condition is successfully met, as TR2 has a lower power rating (1100 VA) and a higher short-circuit voltage (4.23%) compared to TR1 (3.9%).
6. **Utilization coefficient (at least 95%):** Using the formulas from the theoretical background, the maximum safe power transfer before overloading the lower impedance transformer (TR1) was calculated:
- 7.

$$S'_1 = 2300 \cdot \frac{3.9}{3.9} = 2300 \text{ VA}$$

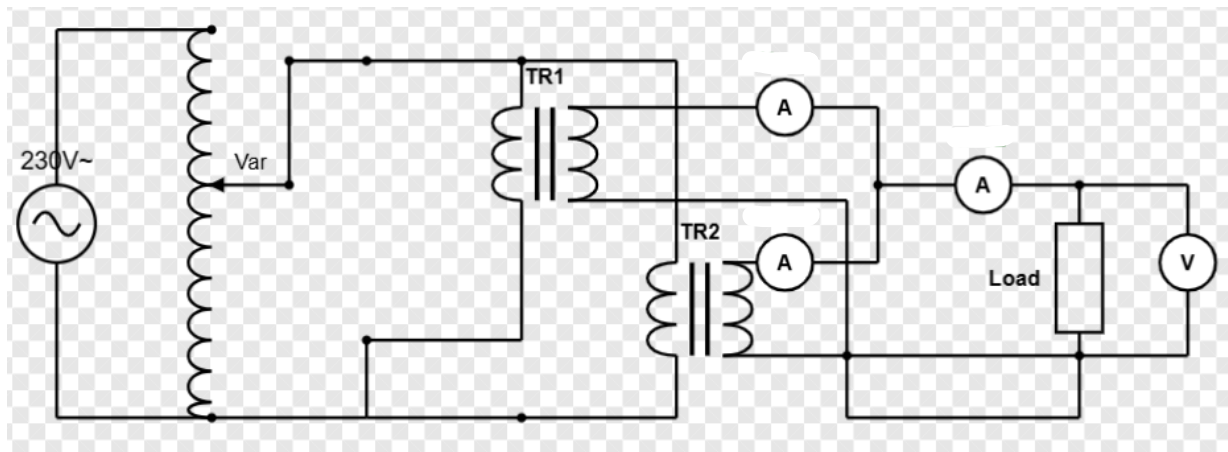
$$S'_2 = 1100 \cdot \frac{3.9}{4.23} = 1014.18 \text{ VA}$$

$$\gamma = \frac{S'_1 + S'_2}{S_{n1} + S_{n2}} = \frac{2300 + 1014.18}{2300 + 1100} = \frac{3314.18}{3400} = 0.975$$

The utilization factor is **97.5%**, which is greater than the required 95%, satisfying the condition.

3. Circuit Diagram

Below is the schematic sketch of the single-phase connection used during the laboratory procedure. A variable autotransformer is placed at the input to regulate the voltage fed to the primary windings of TR1 and TR2. Ammeters are strategically placed to measure the currents of TR1 (I_1), TR2 (I_2), and the total current (I_3) flowing to the resistive load.



The circuit was energized, and measurements were taken at three different load levels by adjusting the resistive load and ensuring a steady input utilizing the autotransformer.

4. Experimental Measurements

Note on Ammeter Reading: For TR1 (I_1), the ammeter constant was calculated as

$$c_1 = \frac{2.5 \text{ A}}{100 \text{ div}} = 0.025 \text{ A/div}$$

For example, a reading of 45 divc *orresponds to* $I_1 = 45 \cdot 0.025 = 1.125 \text{ A}$.

Measurement No.	Voltage V	Current TR1 I_1	Current TR1 I_2	Total Current I_3
1	61	1.125	0.45	1.60
2	99	1.825	0.80	2.60
3	122	2.300	1.00	3.25

5. Theoretical vs. Experimental Power Distribution

As requested, the theoretical values of the apparent power carried by each transformer (S_1 and S_2) were calculated based on the total measured power S_3 , using the power distribution ratio formula from Example 2.6.

$$\frac{S_{TR1}}{S_{TR2}} = \frac{\frac{S_{n1}}{u_{k1}}}{\frac{S_{n2}}{u_{k2}}} = \frac{\frac{2300}{0.039}}{\frac{1100}{0.0423}} = \frac{58974.36}{26004.73} \approx 2.2678$$

From this ratio, we establish that:

$$S_{TR1} = 2.2678 \cdot S_{TR2}$$

Since the total power is $S_3 = S_{TR1} + S_{TR2}$, we can substitute to find the theoretical fractions:

$$S_3 = 2.2678 \cdot S_{TR2} + S_{TR2} = 3.2678 \cdot S_{TR2}$$

$$S_{TR2(Theoretical)} = \frac{S_3}{3.2678}$$

$$S_{TR1(Theoretical)} = S_3 - S_{TR2(Theoretical)}$$

Measurement 1 Calculation

- **Total Measured Power:** $S_3 = V \cdot I_3 = 61 \text{ V} \cdot 1.6 \text{ A} = 97.6 \text{ VA}$
- **Theoretical S_2 :** $S_2 = 97.6 / 3.2678 = 29.87 \text{ VA}$
- **Theoretical S_1 :** $S_1 = 97.6 - 29.87 = 67.73 \text{ VA}$
- **Experimental S_2 :** $61 \text{ V} \cdot 1.125 \text{ A} = 68.63 \text{ VA}$
- **Experimental S_1 :** $61 \text{ V} \cdot 0.45 \text{ A} = 27.45 \text{ VA}$

Measurement 2 Calculation

- **Total Measured Power:** $S_3 = V \cdot I_3 = 99 \text{ V} \cdot 2.6 \text{ A} = 257.4 \text{ VA}$
- **Theoretical S_2 :** $S_2 = 257.4/3.2678 = 78.77 \text{ VA}$
- **Theoretical S_1 :** $S_1 = 257.4 - 78.77 = 178.63 \text{ VA}$
- **Experimental S_2 :** $99 \text{ V} \cdot 0.80 \text{ A} = 79.20 \text{ VA}$
- **Experimental S_1 :** $99 \text{ V} \cdot 1.825 \text{ A} = 180.68 \text{ VA}$

Measurement 3 Calculation

- **Total Measured Power:** $S_3 = V \cdot I_3 = 122 \text{ V} \cdot 3.25 \text{ A} = 396.5 \text{ VA}$
- **Theoretical S_2 :** $S_2 = 396.5/3.2678 = 121.34 \text{ VA}$
- **Theoretical S_1 :** $S_1 = 396.5 - 121.34 = 275.16 \text{ VA}$
- **Experimental S_2 :** $122 \text{ V} \cdot 1.00 \text{ A} = 122.00 \text{ VA}$
- **Experimental S_1 :** $122 \text{ V} \cdot 2.30 \text{ A} = 280.60 \text{ VA}$

6. Summary and Conclusion

The laboratory exercise successfully verified the principles of parallel transformer operation. When evaluating the conditions for parallel operation, we found that despite a slight deviation in the power rating ratio limit (2.09:1 instead of 2:1), the transformers were suitable for parallel operation. The fact that the lower-rated transformer (TR2) possessed a higher short-circuit voltage (4.23% vs 3.9%) successfully prevented it from becoming overloaded, resulting in an excellent theoretical utilization factor of 97.5%.

When analysing the load distribution based on Example 2.6, the theoretically calculated apparent powers align very closely with the experimentally measured powers. For instance, in Measurement 3, the theoretical power for TR2 was 121.34 VA, while the measured power was exactly 122.00 VA (an error of only $\approx 0.5\%$). TR1's experimental values were consistently just slightly higher than theoretical projections, but the overall alignment strongly validates the theoretical model of load division being inversely proportional to the short-circuit impedances of the transformers.